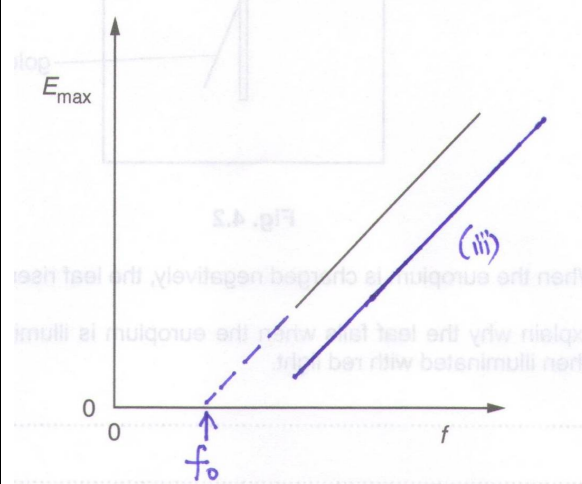


4(a)

Threshold frequency refers the **minimum frequency** of the illuminating electromagnetic radiation source that will cause a photoelectron to be ejected when the electromagnetic radiation source is radiated on a target metal.

4(b)(i)
(iii)



$$hf = hf_0 + E_{\max}$$

$$E_{\max} = hf - hf_0$$

A graph of $E_{\max}$ against f shows a gradient of h .
When $f=f_0$ occurs, $E_{\max} = 0$

4(b)(ii)	Gradient of line equals to Planck's constant.
4(c)(i)	$\frac{hc}{\lambda} = \phi + KE_{\max}$ $\frac{\left(6.63 \times 10^{-34}\right)\left(3.0 \times 10^8\right)}{490 \times 10^{-9}} = 2.5 \times 1.6 \times 10^{-19} + \frac{1}{2}\left(9.11 \times 10^{-31}\right)v^2$ $v = 1.14 \times 10^5 \text{ m s}^{-1}$
4(c)(ii)1.	Energy of photon is inversely proportional to its wavelength. Blue light has a smaller wavelength as compared to red light and thus has higher energy. It is able to overcome the work function of metal and cause electrons to be emitted and hence reduces the force of repulsion between the metal rod and gold leaf and leaf falls.
4(c)(ii)2.	If energy of photon from light is greater than work function of europium, electrons are emitted from metal and thus gold leaf and metal rod will be more positively charged and caused greater deflection. If energy of photon from light is lesser than work function of europium, electrons will not be emitted and repulsive force between metal rod and leaf remains the same and deflection will remain the same.
5(a)	Mass of copper in 1 m^3 of copper = 8960 kg